



NITROGEN USE EFFICIENCY (NUE), N CONCENTRATION, UPTAKE AND YIELD PARAMETERS OF MAIZE AS INFLUENCED BY DIFFERENT NITROGEN FERTILIZER RATES IN NORTHERN GUINEA SAVANNA ALFISOLS

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ABSTRACT

Key words:

Nitrogen Use Efficiency (NUE), maize, Savanna, Alfisols, nitrogen, yields

Maize is a very high nutrients demanding crop, which requires adequate nutrition for maximum performance. The major nutrients required by maize are nitrogen (N), phosphorus (P) and potassium (K) as well as micronutrients for proper growth and development. This study was conducted to investigate the effect of different nitrogen fertilizer rates on nitrogen use efficiency (NUE), N concentration, and uptake and yield parameters of maize. The treatments in this experiment were 4 nitrogen rates (0, 50, 100 and 150 kg/ha) replicated 3 times and laid in Randomized Completely Block Design (RCBD). The NUE parameters measured was recovery and physiological efficiency while the yield parameters were number of cobs, stover yield, grain yield and 1000 grain weight. The data collected showed significant effect of treatment on NUE with 50 kg/ha as the best performing treatment for NUE recovery efficiency while 100 kg/ha for NUE physiological efficiency. The 50, 100 and 150 kg N/ha was found to be statistically the same, but 150 kg N/ha has the highest mean value for all the yield parameters. Result of N concentration and uptake follow similar trend with yield parameters. From this study, it can be concluded that 100 kg N/ha is the optimum for the maize in the study area. The recommended N rates for maize in the study area is 100 kg N/ha

INTRODUCTION

Maize (*Zea Mays* L.) is a very high nutrients demanding crop, which requires adequate nutrition for maximum performance (Rachid *et al.*, 2004). The major nutrients required by maize are nitrogen (N), phosphorus (P) and potassium (K) as well as micronutrients for proper growth and development (Anjorin *et al.*, 2018). Sangoi *et al.* (2001) reported that, maize crop requires up to about 200 kg N/ha, 50 to 80 kg P/ha and 60 to 100 kg K/ha. Supplying nutrients below required quantity reduce yield. Similar reports by FAO (2007), strongly suggest on the necessity of N, P and K for maize, which insufficiency application adversely affects maize yield. Also, study by Sanchez (1997), observed the lack of proper management to supplement nutrients removed by the crop can result in soil nutrient losses of about 22 kg of N, 2.5 kg of P, and 15 kg of K per hectare per year. Cermak and Smananova (2012) also studied the consequences of insufficient nutrient replenishment would render a previously fertile piece of land unproductive in less than 30 years.

Nitrogen is a major component of numerous plant biological compounds that play crucial roles in photosynthetic activities and consequently influence crop yield (Zhao, *et al.*, 2005). Adequate N levels result in dark green leaves, whilst deficiencies cause leaf chlorosis (Tajul

et al., 2013). Nitrogen is required by maize in higher amounts than any other element and maize grain quality and quantity are improved with adequate nitrogen levels in the soil. The most frequent source of nitrogen that would readily be available to plant is through fertilizer application, and its major threats are ease of leaching, volatilization, high cost of procurement, and environmental pollution such as contamination of surface and ground water (Kotschi., 2013). Generally, NH₃ volatilization reported to be a predominant loss mechanism that reaches up to 50% of applied N (Sommer *et al.*, 2004).

Nitrogen use efficiency is the amount of nitrogen a crop takes and retains until harvest, compared to the amount of nitrogen that was actually available for the crop to uptake, with a particular emphasis on how much fertilizer is applied to soils (Cameron *et al.*, 2013). Universally, the nitrogen use efficiency for maize is approximately 33%, with estimated averages of 29% and 42% for the developing and developed countries respectively (Walsh *et al.*, 2012). Although it is impossible to achieve 100% NUE in any crop production system; however, improving crop production systems through management of fertilizer as well as soil- and plant-water relationships could result in improved NUE up to 50% or above in cropping systems. Hence, applying the optimum N level is the most viable way to increase yield of crop such as maize and improving efficiency in the use of nitrogen.

MATERIALS AND METHODS

The Study Area

The study was conducted at the Experimental Farm of the Institute for Agricultural Research (IAR), Samaru, during the 2020 rainy season. Samaru is located at the geographic coordinates of latitude 11° 11'N and longitude 07° 38'E, with an altitude of 686 m above sea level in the northern guinea savanna ecological zone of Nigeria. The climate of the study area is characterized by the occurrence of distinct wet and dry seasons (Oluwasemire and Alabi, 2004). The area has a mean annual rainfall of about 1011 ±1610 mm, with an average intensity of 50 mm/hr. A mean annual temperature range of about 23°C has also been recorded for the area (IAR Met. Unit, 2017). The annual rainfall attains its highest value in August and drops rapidly to its lowest in October.

Treatments and experimental design

The treatments consisted of 4 rates of nitrogen 0, 50, 100 and 150 kgN/ha replicated 3 times in randomized completely block design.

Cultural practices

The test crop was hybrid maize variety Oba-super 2 planted at spacing of 75 cm between and 25 cm within row. The total population obtained by this spacing was 53,333 plants ha⁻¹. The plot size was 5 m x 4.5 m. Space of 1 m was giving between each plots. Urea was split applied at 2 and 6 weeks after planting (WAS) using basal application method, while muriate of potash and single super phosphate were applied with first split dose of N at 2 WAS. The thinning was done at two weeks after sowing (2 WAS). The plants were thinned to one stand and the thinned plants were replanted where there was no germination. Harvesting was done at maturity at about 112 days after sowing. The harvesting was done manually with knife by cutting the cobs and the operation was carried out when the cobs had turned brown and dried. Threshing was done after the harvested cobs were properly air dried.

Nitrogen Use Efficiency

Nitrogen use efficiency is the amount of nitrogen a crop takes and retains until harvest, compared to the amount of nitrogen that was actually available for the crop to uptake, with a particular emphasis on how much fertilizer is applied to soils (Cameron *et al.*, 2013).

Plant tissue sampling and preparation

The two index leaf and upper portion of stem of the plants were sampled from each plot. Two plants per stand were selected and the samples collected were put into an envelope. The envelope containing plant tissue samples was labeled and dried in an oven for about 48 hours at 65°C. After drying, the plant sample was ground by the grinding machine and put in a labeled polyethene bag and kept for analysis. From the samples kept, analysis of total nitrogen of was carried out by micro- Kjeldahl distillation procedure according to (Bremmer, 1996).

N concentration determination

The analysis of total nitrogen was carried out micro- Kjeldahl distillation procedure according to (Bremmer, 1996). The total nitrogen determined represents the N concentration of the plant.

Nitrogen uptake determination

The nitrogen uptake is amount of nitrogen absorbed by the plant. The uptake was determined according to Satter, (2015) formulae

$$\text{Uptake (kg/ha)} = \text{N concentration (g/kg)} \times \text{grain yield (kg/ha)}$$

Calculation of nitrogen use efficiency

The nitrogen use efficiency of the plants from each plot were obtained by subjecting the total nitrogen concentration obtained from the analysis of the plant tissue to specialized calculations by using various Nitrogen Use Efficiency (NUE) formulae as shown below

Physiological efficiency

Physiological efficiency refers to the ability of a plant to transform a given amount of acquired nitrogen into grain yield. It refers to the grain yield per unit nitrogen uptake. It can be calculated as follows. (Satter, 2015)

$$PE = \frac{(Y_{NA} - Y_{N0})}{(NU_{NA} - NU_{N0})}$$

Where:

PE = Physiological efficiency

Y_{NA} = yield (kg/ha) with nitrogen addition

Y_{N0} = yield (kg/ha) without nitrogen addition

NU_{NA} = nitrogen uptake (kg/ha) with nutrient addition

NU_{N0} = nitrogen uptake (kg/ha) without nutrient addition

Recovery efficiency

Recovery efficiency refers to the increase in nitrogen uptake by plant per unit applied nitrogen. It can be calculated as follows (Satter, 2015).

$$RE = \frac{(NU_{NA} - NU_{N0})}{NU_{RN}}$$

Refer:

RE = recovery efficiency

NU_{NA} = nitrogen uptake (kg/ha) with nitrogen addition
 NU_{N0} = nitrogen uptake (kg/ha) without nitrogen addition
 NU_{RN} = rate of nitrogen addition (kg/ha)

Post- harvest data:

Number of cobs: total number of cobs obtained from each plot was recorded

Stover yield: the dry matter (Stover) yield obtained from each plot was recorded in kilo gram. It was done by measuring remaining portion of the plant after cobs were removed. The measurement was done with beam balance after drying.

Grain yield (tha^{-1}): The air-dried cobs were threshed, cleaned and weight of grain recorded on the basis of grain yield per n et plot. Grain yield per ha was calculated and expressed in tonnes per hectare.

1000-grain weight (g): The weight of thousand grains (g) was recorded from grain samples drawn from the produce obtained from each of the plot.

Statistical Analysis

The Statistical analysis of all parameters collected from the field on plants was subjected to Analysis of Variance (ANOVA) using SAS package (SAS, 1999). The means were separated using the Duncan Multiple Range (DMRT) at 5% level of probability.

RESULT AND DISCUSSION

Effect of Nitrogen Rates on Nitrogen Use Efficiency (NUE) in maize

The nitrogen use efficiency (NUE) was express as the recovery efficiency and physiological efficiency in this study (Table 1). The effect of N application and rates on physiological efficiency (PE) was highly significant. Values increased with increase in N rates up to 100 kg N/ha, with a value of 52.2%. The 0 kg N/ha was least with 0.121% representing indigenous N factor, strongly suggesting the need for nitrogen fertilization of the soils for high and sustained yield of maize. The PE declined at the higher rate of 150 kg N/ha. This result is in agreement with that of (Cameron *et al.*, 2013). The effect of N rates on nitrogen recovery (RE) efficiency was significant and was found to decrease with increase in N applied with highest mean in 50 kg N/ha (39.7%) and least mean at 150 kg N/ha (17.86 %).

Table 1: Effects of N rates on Nitrogen Use Efficiency (NUE) of maize

Treatments	Nitrogen Use Efficiency (NUE)	
	Recovery Efficiency (%)	Physiological Efficiency (%)
<u>N rate (kg/ha)</u>		
0	00.00 ^c	0.12 ^c
50	39..70 ^a	2.69 ^{ab}
100	22.49 ^b	5.22 ^a
150	17.86 ^b	4.89 ^a
Mean	20.01	0.91
S.E±	2.59*	0.98**

Means follow by the same letter(s) in the same column are not significantly different at 5% level of probability, NS= *=significance, **=highly significance

consequence will be high leaching losses and possible ground water contamination at high rates of N application. Generally, RE obtained in this study ranged from 15.72 to 39.7% across all factors considered. The 50 kg N/ha plots gave the highest value of 39.7%. These results are in agreement research finding of Cameron *et al.* (2013) who reported a decrease in NUE with increasing nitrogen level, especially when it was applied in excess of the crop requirement.

Effect of Nitrogen Rates on Maize Yield Parameters

The effect of nitrogen application and rates on number of cobs were also highly significant ($p>0.01$) and increased to the highest rates of 150 kg N/ha applied (Table 2). However, the practices of N fertilization alone stimulated similar response, as mean values for the N rates were not significantly different, but were all superior to the N control (0 kg N/ha) plot by an

Table 2: Effects of N rates on yield parameters of maize

Treatments	Number of combs	1000 Grain Weight (g)	Stover Yield (kg/ha)	Grain Yield (kg/ha)
<u>N rate (kg/ha)</u>				
0	46.89 ^b	234 ^b	4003.2 ^b	2273.1 ^c
50	56.41 ^a	277 ^a	5184.0 ^a	3027.2 ^b
100	57.33 ^a	291 ^a	5228.1 ^a	3309.9 ^{ab}
150	59.11 ^a	298 ^a	6063.5 ^a	3580.1 ^a
Mean	54.96	275	5119	3047.6
S.E±	4.93**	239**	409	161**

Means follow by the same letter(s) in the same column are not significantly different at 5% level of probability, NS= *=significance, **=highly significance

average of 21%. Though this parameter can be said to be strongly controlled by the genetic factor, it can also be strongly enhanced by high level of fertilization and faster growth response. This result agrees with that of Tanimu *et al.* (2007) and Wang *et al.* (2015) who found an increased in grain yield and yield related components of maize with N rates application compared with control plots. The effects of N rates on stover yield were highly significant, and response to applied N was significant up to the highest rate of N applied (150 kg/ha N/ha), and mean values at 100 kg/ha and 150 kg N/ha were significantly different. Stover yield increased consistently with increase in N rates. The result agrees with finding of Zerihun and Hailu (2017). Response of 1000 grain weight to N application increased with increase in N rates to the highest level tested (150 kg/ha). All N treatment plots were significantly different from the N control plots, but differences were not statistically significant across N rates. The high grain weight found in 150 (kg/ha) is in line with the finding of Zerihun and Hailu (2017). The effects of N application and rates on

grain yield were highly significant, and yield values increased to the highest rate of N applied (3580 kg/ha at 150kg/ha). Significant response of grain yield to nitrogen rates was only up to 100 kg N/ha and was not significantly different from 150 kg N/ha. The lowest yield values were obtained at 0 kg N/ha treatment (2273 kg/ha). This result agrees with finding of Zerihun and Hailu (2017).

Effect of Nitrogen Rates on Nitrogen Concentration and Uptake in Maize

Variation in N concentration with N rates increased only up to 50 kg N/ha and declined thereafter (Table 3). This further lends support to the explanation that tissue dilution strongly controlled the results found. As shown in Table 2, stover yield increased with increase in N rates which explained the decreasing N concentration with increasing biomass yield (stover + grain), that resulted in nutrient dilution in the plant at higher yields. This is in consonance with the findings of Zhenling *et al.* (2009) who studied the minimum N concentration associated with high performing treatments under different maize cultivars. The effect of N rates on N uptake by maize was highly significant ($p>0.01$) and increased with increased rates of N applied. N uptake values ranged from 72 kg/ha for the 0 kg N/ha plots to 121kg N/ha for the 150 kg N/ha treatment. N uptake increased consistently with rate of N applied as would be expected. Moll *et al.* (1982) observed similar findings on significant differences among maize cultivars in N uptake after the ten-leaf stage

CONCLUSION AND RECOMMENDATION

Yield component of maize determined in this study (number of cobs, 1000 grain weight and stover yield) were duly influenced maximally by the 100 kg N/ha (3825 kg/ha). With these yield components, mean N response was only up to 100 kg N/ha for all parameters. NUE, expressed as Recovery Efficiency (RE), was low and this was attributed to effects of leaching and surface run-off. Best Recovery Efficiency (RE) value results were obtained with 50 kg N/ha (39.7% RE). RE decreased with increase in N rate applied. From foregoing it can be recommended that, practices that would enhance surface run-off and low infiltration rate such as over fertilization and odd timing application should be discouraged to ensure improvement in NUE and maize yield.

Table 3: Effects of N rates on N concentration and Uptake in maize

Treatments	N concentration (g/kg)	Uptake(kg/ha)
<u>N rate (kg/ha)</u>		
0	1.13 ^b	7242.8 ^a
50	1.27 ^a	10659.3 ^a
100	1.25 ^a	10681.1 ^b
150	1.25 ^a	12102.6 ^a
Mean	1.22	10171.5
S.E±	0.03*	706**

Means follow by the same letter(s) in the same column are not significantly different at 5% level of probability, NS= *=significance, **=highly significance

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